COMP547 Deep Unsupervised Learning

Spring 2026

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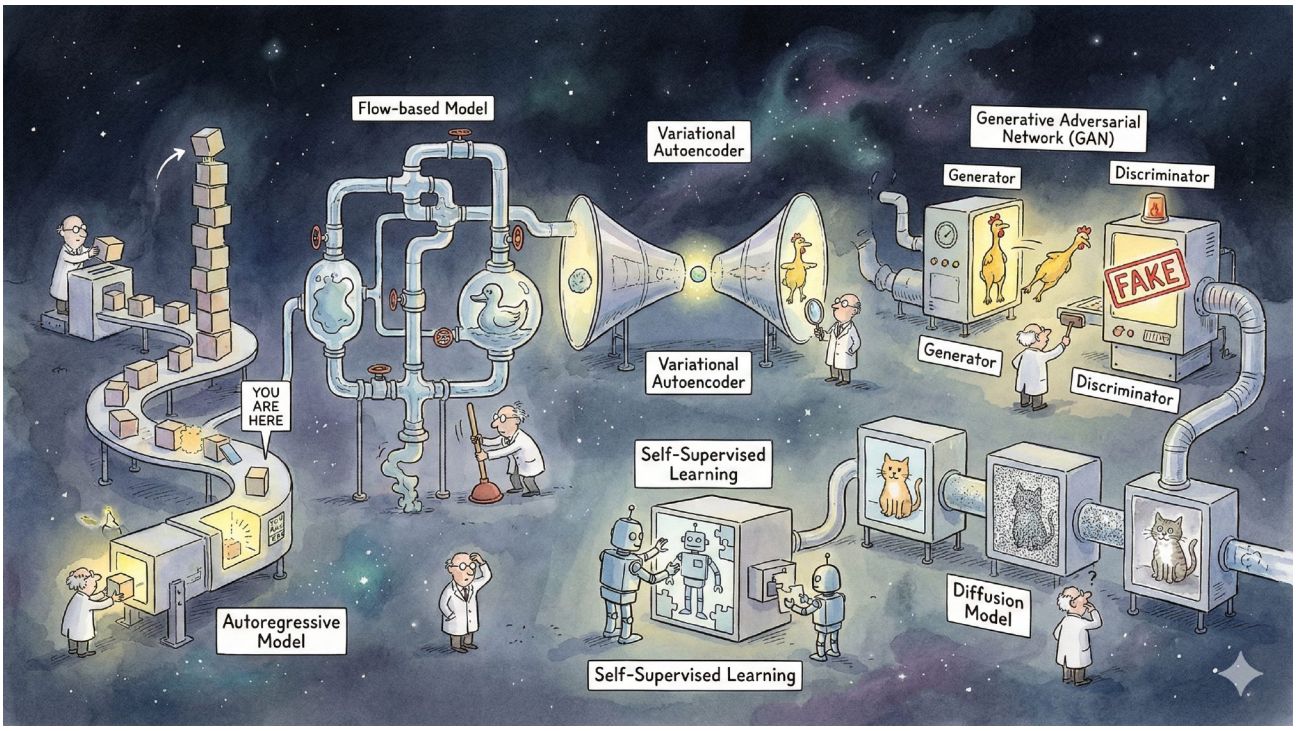
Lectures: Tuesday and Thursday 14:30-15:40 (SOS 103)

Tutorials: Friday 14:30-15:40 (SOS Z27)



An illustration generated by Gemini Nano Banana Pro using the prompt "Can you generate a wide-screen, funny illustration in the style of Jean-Jacques Sempe in a dark background for my deep unsupervised learning class. It includes generative models, namely autoregressive models (like you), flow-based models, variational autoencoders, generative adversarial models, and diffusion models, and self-supervised learning. Don't use any French."

COURSE DESCRIPTION

This course provides a thorough and comprehensive exploration of the fundamental concepts and recent advances in deep unsupervised learning. The curriculum covers a range of deep generative models, including autoregressive models, normalizing flows, variational autoencoders, generative adversarial networks, and diffusion models. The course concludes with an in-depth examination of self-supervised learning techniques.

Prerequisites: COMP547 is intended for graduate students enrolled in Computer Science MS and PhD programs. Senior undergraduate students and all non-COMP graduate students need the instructor's permission to register for the class. The prerequisites are[1]:

- Good programming skills (for practicals and the course project)
- Calculus (differentiation, chain rule) (MATH203) and Linear Algebra (vectors, matrices, eigenvalues/vectors) (**MATH107**)
- Basic Probability and Statistics (random variables, expectations, multivariate Gaussians, Bayes rule, conditional probabilities) (**ENGR200**)
- Machine Learning (ENGR421) or Deep Learning (COMP441/541)
- Optimization (cost functions, taking gradients, regularization)

Each student must complete a self- quiz on prerequisite math knowledge.

[1] Courses in bold are official prerequisites.

Grading Policy: The grading will be based on 3 assignments (21%, 7% each), midterm exam (10%), course project (41%), paper presentations (8%), paper reviews (10%), and class participation (10%).

Webpage: `https://aykuterdem.github.io/classes/comp547.s25`

Schedule:

W	*Date*	*Topic*	*Assignments*
1	2/10-12	Introduction, Spatial Processing with CNNs	
2	2/17-19	Sequential Processing with NNs, Attention and Transformers	
3	2/24-26	Autoregressive Models	Assg 1 out
4	3/3-5	Normalizing Flow Models	
5	3/10-12	Latent Variable Models	Assg 1 due, Assg 2 out
6	3/17-19	*Spring Break*	
7	3/24-26	Generative Adversarial Networks I	*Project proposal due*
8	3/31-4/2	Generative Adversarial Networks II	Assg 2 due, Assg 3 out
9	4/7-9	Diffusion Models I	
10	4/14-16	Diffusion Models II	
11	4/21	Strengths and Weaknesses of Current Generative Models	Assg 3 due
12	4/28-30	*Project Progress Presentations*	*Project progress reports due*
13	5/5-7	Video Generation	
14	5/12-14	Self-Supervised Learning I	
15	5/21	Self-Supervised Learning II	*Midterm exam*
16	6/9-11	Final Project Presentations	*Final project reports due*

Communication: The course webpage will be updated regularly throughout the semester with lecture notes, programming and reading assignments and important deadlines. All other communications will be carried out through KUHub Learn.

Policies: All work on assignments has to be done individually. The course project, however, can be done in pairs. You are encouraged to discuss with your classmates about the given assignments, but these discussions should be carried out in an abstract way. In short, turning in someone else's work, in whole or in part, as your own will be considered as *a violation of academic integrity.* Please note that the former condition also holds for the material found on the web as everything on the web has been written by someone else.